1

Sets

Chapter Learning Outcomes

CLO

1

- Represent a set using set-builder notation and roster method.
- Understand the numerical sets.
- Understand the concepts: universal set, empty set, finite and infinite Sets, set of sets.
- Describe memberships of sets, including the empty set, using proper notation, and decide whether given items are members and determine the cardinality of a given set.
- Describe the relations between sets regarding membership, equality, subset, and proper subset, using proper notation.
- Perform the operations of union, intersection, complement, and difference, complements, Cartesian products, and power sets.
- Determine whether two sets are equal by determining whether each is a subset of the other.
- Be able to draw and interpret Venn diagrams of set relations and operations and use Venn diagrams to solve problems.
- Calculate and describe the power set for small sets.
- Understand set properties.

Sets

A **set** is unordered collection of objects or elements.

- The elements can be numbers, letters, etc.
- Elements are listed inside curly brackets { }.
- Usually we denote sets with upper-case letters, elements with lower-case letters.

EXAMPLE

A: is a set of integer between 1 and 9.

B: is the set of primary colors.

Describing a Set

1- Roster Method

In this method a set is represented by listing all its elements, separating them by commas and enclosing them in curly bracket

Set of all vowels in the English alphabet:

Set of all odd positive integers less than 10:

Set of all positive integers less than 100:

Set of all integers less than 0:

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Describing a Set

2- Set Builder Notation

In this method, specify the property (or properties) that all members must satisfy.

$S = \{1, 2, 3, \dots, 99\}$

$S = \{x : x \text{ is a positive integer less than } 100\}$

$C = \{1, 3, 5, 7, 9\}$

$C = \{x : x \text{ is an odd positive integer less than } 10\}$

$V = \{a, e, i, o, u\}$

$V = \{a : a \text{ is an English vowel alphabet}\}$

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Finite and Infinite Sets

if a **set** has a starting point and an ending point both, it is a **finite set**, but it is **infinite** if it has no end from any side or both sides.

EXAMPLE

Set	Finite	Infinite

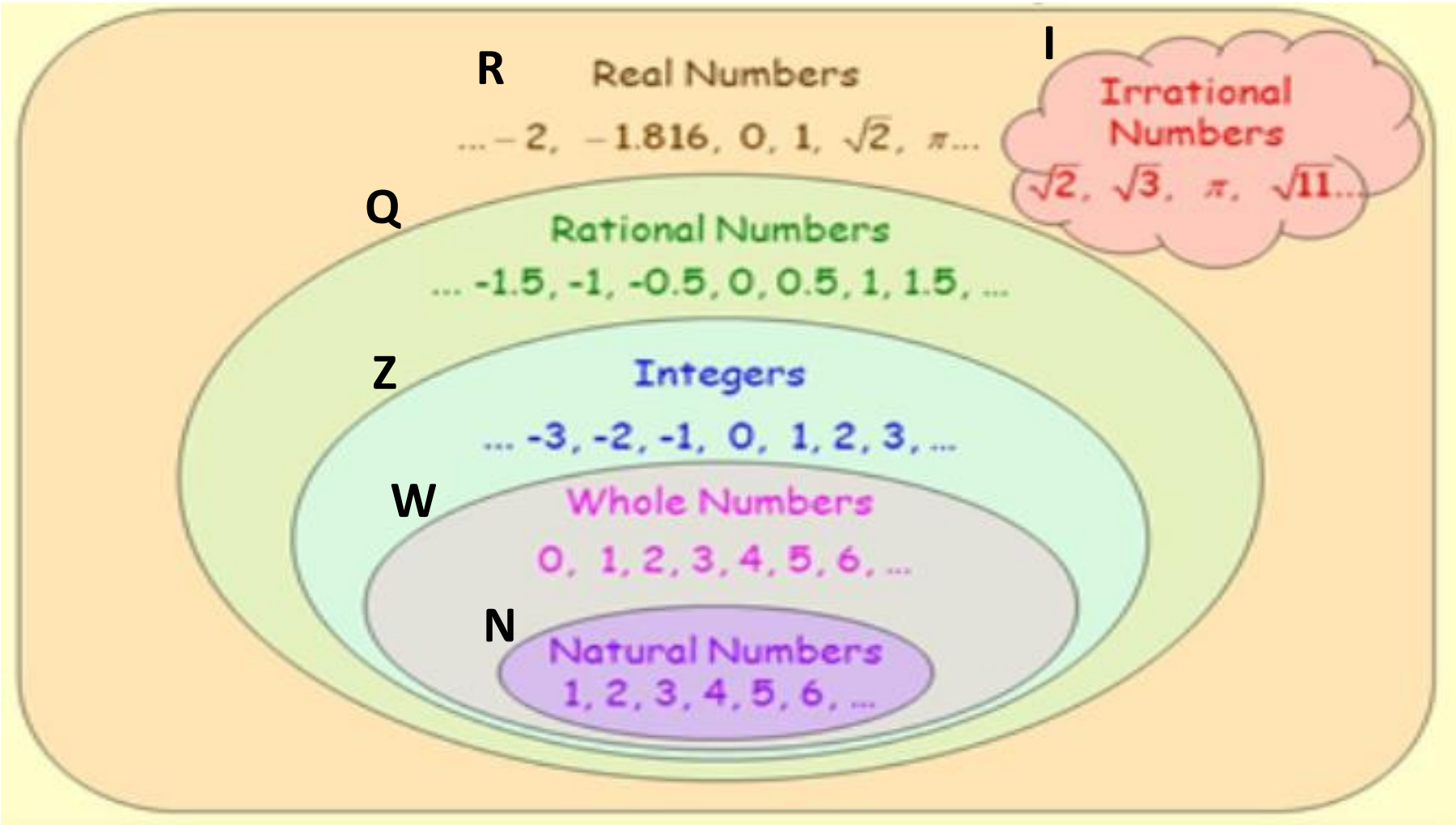
$S = \{1, 2, 3, \ldots, 99\}$

$C = \{1, 3, 5, 7, \ldots\}$

$A = \{\ldots, -3, -2, -1\}$

$B = \{1, 2, 3, 4, \ldots\}$

Numerical Sets



Rational and Irrational Numbers

What is the difference between rational and irrational numbers?

SOLUTION

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Numerical Sets

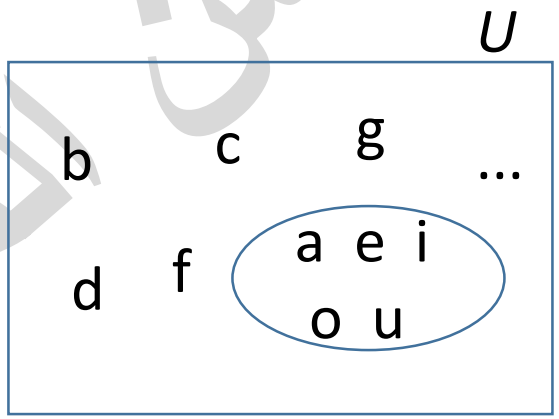
QUIZ

Classify each number as a rational or an irrational number.

1. $\sqrt{4}$
2. 27.5
3. $\sqrt{37}$
4. $1/9$
5. 0.09
6. 0.895210351
7. 0.285714
8. $\sqrt{64}$
9. $\sqrt{2}$
10. 11.5

Universal Set

- The *universal set U* contains everything in the *domain*.
- It's a set that contains everything.
- Well, not *exactly* everything. **Everything that is relevant to our question.**



Venn Diagram
(the domain is the
small letters)

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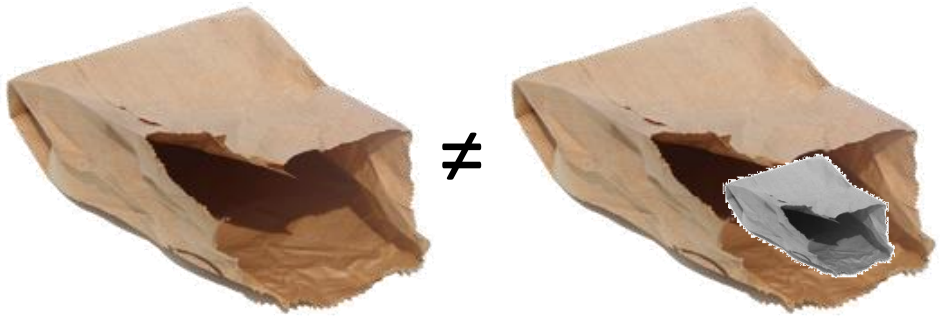
Empty Set

- The empty set has no elements, written as $\emptyset$, or $\{\}$.

$$S = \{x : x \text{ is a prime number between } 14 \text{ and } 16\}$$

$$S = \{\} \text{ or } S = \emptyset$$

- The empty set is different from a set containing the empty set.
- $\emptyset \neq \{\emptyset\}$



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Set Equality

Two sets are equal if and only if they have the same elements.

$\{1,3,5\} = \{3,1,5\}$

$\{1,3,1,1,5,5,3,3\} = \{1,3,5\}$

$\{1,3,3,5\} \neq \{1,1,5\}$

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Exercise For the Student

Which of the following pairs of sets are equal? Justify your answer.

$$(i)\quad A = \{x : x \text{ is a letter of the word "LOYAL"}\},$$
$$B = \{x : x \text{ is a letter of the word "ALLOY"}\}$$

$$(ii)\quad A = \{x : x \in Z \text{ and } x^2 \leq 8\},$$
$$B = \{x : x \in R \text{ and } x^2 - 4x + 3 = 0\}$$

Set of Sets

A **set of sets** is a set, whose elements are themselves all sets.

$$A = \{\{1,2\}, \{3,4\},\{5\}\}$$

EXAMPLE

Let A={1, 2, 3, 4, 5} and B={{1,2}, {3,4}, {5}}.

1. **How many elements does A have?**
2. **How many elements does B have?**
3. **Does A = B?**

Set Cardinality (Size)

The **cardinality** of a set *A*, is the number of elements in *A* and is denoted by $|A|$, or n(A).

EXAMPLE

1. $|\emptyset| =$
2. $|\{1,2,3\}| =$
3. $|\{\emptyset\}| =$
4. $|Z| =$
5. $|\{\{1,2\},\{3,4\},\{5\}\}| =$

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Set Cardinality

EXAMPLE

Find the cardinality of the following sets:

1. $A=\{23,24,\ldots,37,38\}$.
2. $B=\{1,\{2,3,4\},\emptyset\}$.
3. $C=\{x\in Z : -2\leq x\leq 100\}$.
4. $D=\{3,3,3,3\}$.

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Set Membership

The following notation is used to show set membership

$x \in A$ means that x is a member of the set A

$x \notin A$ means that x is not a member of the set A.

Examples:

1. $x \in \{x, y, z\}$
2. $w \notin \{x, y, z\}$
3. $\{1, 2\} \in \{\{1, 2\}, \{1, 2, 3\}\}$.
4. $\{1, 3\} \notin \{\{1, 2\}, \{1, 2, 3\}\}$.

Subset and Proper Subset

A is a **subset** of B $A \subseteq B$

If every member of set A is also a member of set B, then A is a subset of B. May be A = B.

A is a **proper subset** of B $A \subset B$

If every member of set A is also a member of set B and $A \neq B$ then A is a proper subset of B.

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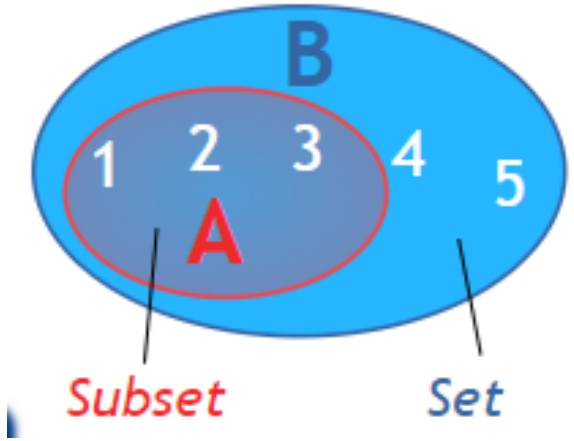
Subset and Proper Subset

$B = \{1, 2, 3, 4, 5\}$

$A = \{1, 2, 3\}$

A is a subset of B

$A \subseteq B$



Also

A is a proper subset of B

$A \subset B$

Notice that when A is a proper subset of B then it is also a subset of B.

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Subset and Proper Subset

Example:

{1, 2, 3} is a **subset** of {1, 2, 3}, but is **not** a **proper subset** of {1, 2, 3}.

Example:

{1, 2, 3} **is** a **proper subset** of {1, 2, 3, 4} because the element 4 is not in the first set.

Example

Let A={1,2,3,4,5,6}, B={2,4,6}, C={1,2,3} and D={7,8,9}. Determine which of the following are true, false, or meaningless.

1. $A \subset B$.
2. $B \subset A$.
3. $B \in C$.
4. $\emptyset \in A$.
5. $\emptyset \subset A$.
6. $A < D$.
7. $3 \in C$.
8. $3 \subset C$.
9. $\{3\} \subset C$.

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Exercise For the Student

Let A = {1, 2, {3, 4}, 5}. Which of the following statements are incorrect and why?

$(i)\ \{3, 4\} \subset A$ $\quad (ii)\ \{3, 4\} \in A$ $\quad (iii)\ \{\{3, 4\}\} \subset A$

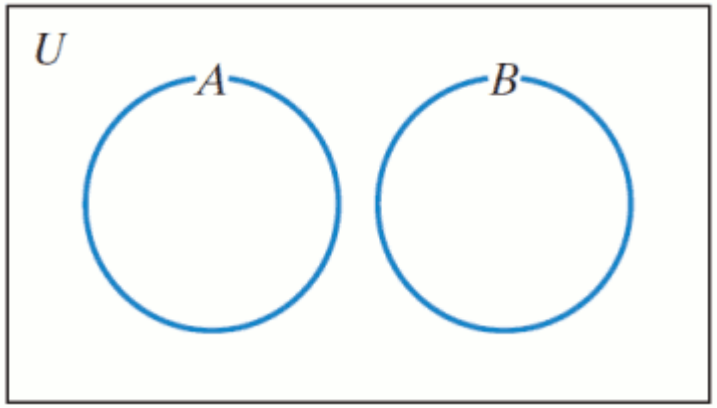
$(iv)\ 1 \in A$ $\quad (v)\ 1 \subset A$ $\quad (vi)\ \{1, 2, 5\} \subset A$

$(vii)\ \{1, 2, 5\} \in A$ $\quad (viii)\ \phi \subset A$ $\quad (ix)\ \phi \in A$

$(x)\ \{\phi\} \subset A$

Disjoint Sets (Mutually Exclusive)

Two sets that have no elements in common are called **disjoint sets**.



EXAMPLE

Determine if the following sets are disjoints or not:

A = {1, 2, 3} and B = {2, 4, 6}

A = {1, 3, 5} and B = {2, 4, 6}

Power Sets

Definition: The set of all subsets of a set A, denoted $\mathcal{P}(A)$, is called the *power set* of A.

If a set has n elements, then the cardinality of the power set is 2^n.

EXAMPLE

Determine the power set for the following set:

$A = \{1, 2\}$

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Power Sets

EXAMPLE

1- If C={1,2,3}, what is the cardinality of $\mathcal{P}(C)$?

2- If A = {a,b}, find the power set of A.

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Power Sets

EXAMPLE

3- If B={ø,{ø}}, find the power set of B.

2- If D = ø, find the power set of D.

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Cartesian Product

Definition: The *Cartesian Product* of two sets *A* and *B*, denoted by $A \times B$ is the set of ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

EXAMPLE

If A = {a, b} B = {1, 2, 3}, find A × B and | A × B|.

Cartesian Product

EXAMPLE

What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

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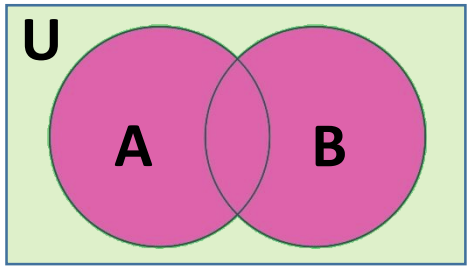
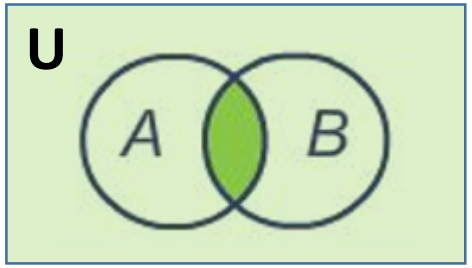
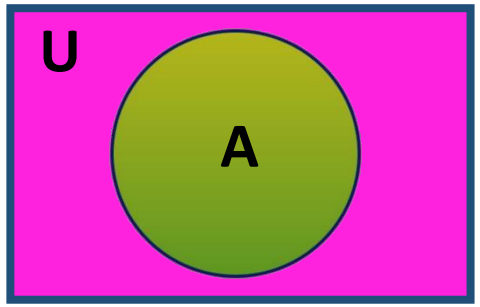
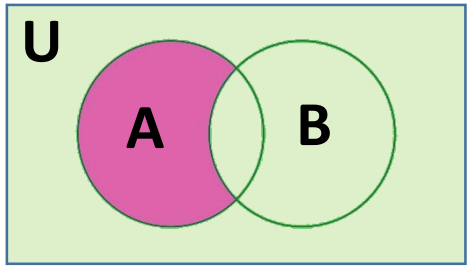
Set Operations

	Notation	Keyword	Venn Diagram
1- Union	$A \cup B$	or	
2- Intersection	$A \cap B$	and	
3- Complement	A' A^c $\overline{A}$	not	

Set Operations

	Notation	Keyword	Venn Diagram
4- Difference	A - B	minus	

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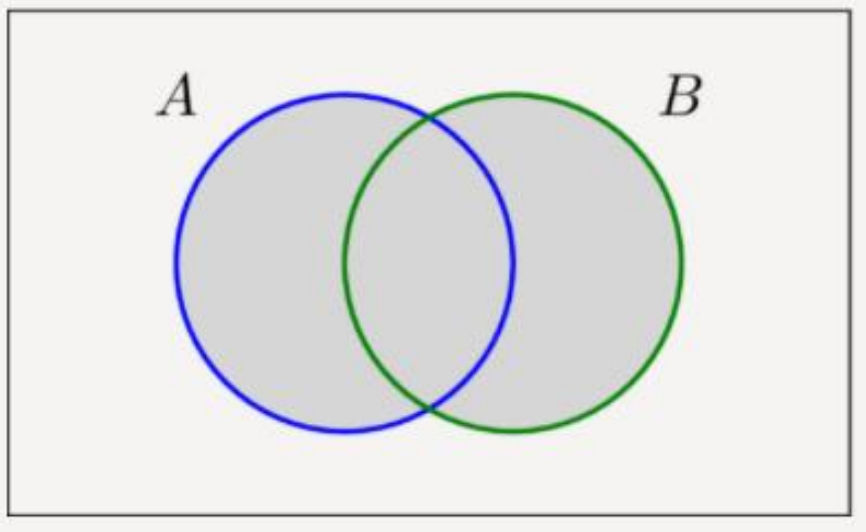
Union

DEFINITION OF THE UNION OF SETS

The **union** of sets A and B, written $A \cup B$, is the set of elements that are members of set A or of set B or of both sets. This definition can be expressed in set-builder notation as follows:

$$A \cup B = \{x \mid x \in A \quad \text{or} \quad x \in B\}.$$



Union

Find each of the following unions:

a. $\{7, 8, 9, 10, 11\} \cup \{6, 8, 10, 12\}$

b. $\{1, 3, 5, 7, 9\} \cup \{2, 4, 6, 8\}$

c. $\{1, 3, 5, 7, 9\} \cup \emptyset$

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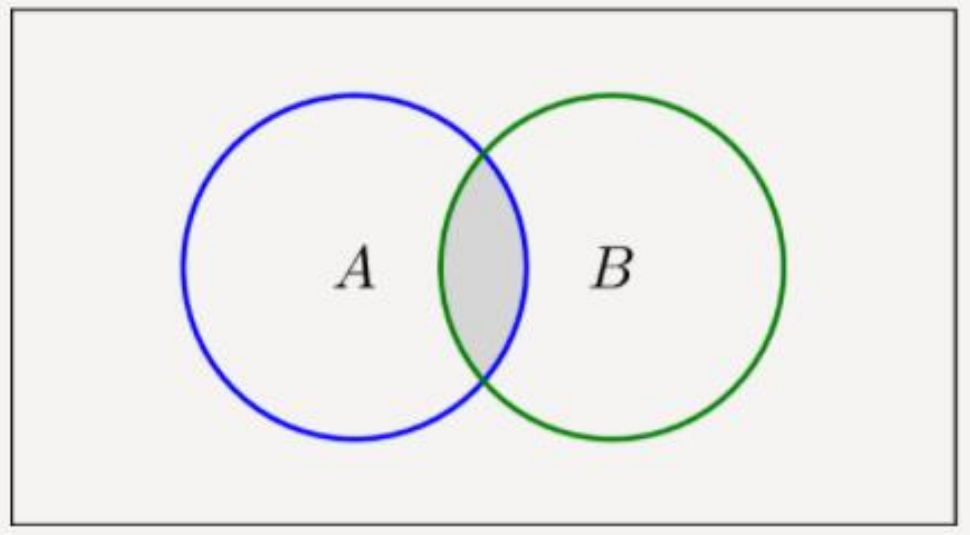
Intersection

DEFINITION OF THE INTERSECTION OF SETS

The **intersection** of sets A and B, written $A \cap B$, is the set of elements common to both set A and set B. This definition can be expressed in set-builder notation as follows:

$$A \cap B = \{x \mid x \in A \quad \text{and} \quad x \in B\}.$$



Intersection

Find each of the following intersections:

a. $\{7, 8, 9, 10, 11\} \cap \{6, 8, 10, 12\}$

b. $\{1, 3, 5, 7, 9\} \cap \{2, 4, 6, 8\}$

c. $\{1, 3, 5, 7, 9\} \cap \emptyset$

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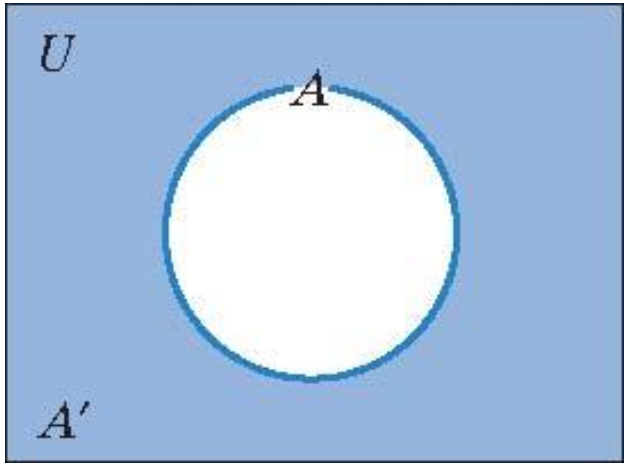
Complement

DEFINITION OF THE COMPLEMENT OF A SET

The **complement** of set A, symbolized by A', is the set of all elements in the universal set that are ***not*** in A. This idea can be expressed in set-builder notation as follows:

$$A' = \{x | x \in U \text{ and } x \notin A\}.$$

The shaded region represents the complement of set A or A'. This region lies outside the circle.



Complement

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $A = \{1, 3, 4, 7\}$. Find A'.

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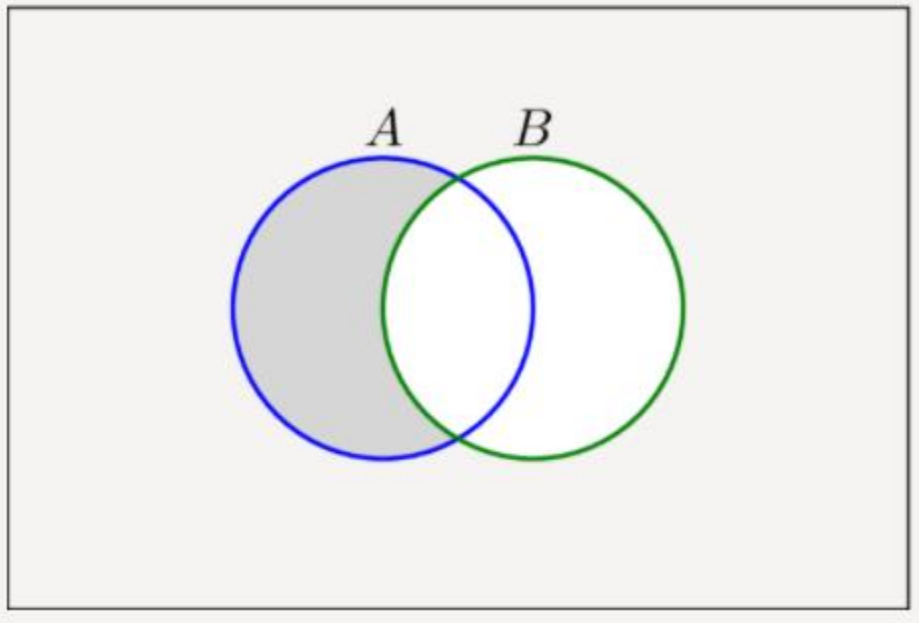
Difference

Definition:

The difference of set A and set B, denoted by $A - B$, is the set that contains exactly all elements in A but not in B.

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}.$$



Difference

Find each of the following differences:

a. {7, 8, 9, 10, 11} - {6, 8, 10, 12}

b. {1, 3, 5, 7, 9} - {2, 4, 6, 8}

c. {1, 3, 5, 7, 9} – {3, 5, 1, 7, 9, 11}

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- $A \cup B$
- $A \cap B$
- $A - B$
- $B - A$
- A^c
- B^c

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- $A \cup B$

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- $A \cap B$

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- A – B

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- B – A

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- A^c

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Example

Let U = R, A = [1, 3] and B = (2, 4). Find:

- B^c

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Properties of sets

$A \cap A = A$ $\quad$ $A \cup A = A$

$A \cap \varnothing = \varnothing$ $\quad$ $A \cup \varnothing = A$

$A \cap B = B \cap A$

$A \cup B = B \cup A$ (commutative)

$(A \cap B) \cap C = A \cap (B \cap C)$

$(A \cup B) \cup C = A \cup (B \cup C)$ (associative)

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Properties of sets

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

(distributive)

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

(De Morgan laws)

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Example

Given:

$U = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{ 1, 3, 7, 9 \}$

$B = \{ 3, 7, 8, 10 \}$

Find

a. $(A \cup B)'$

Solution

Example

Given:

$U = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{ 1, 3, 7, 9 \}$

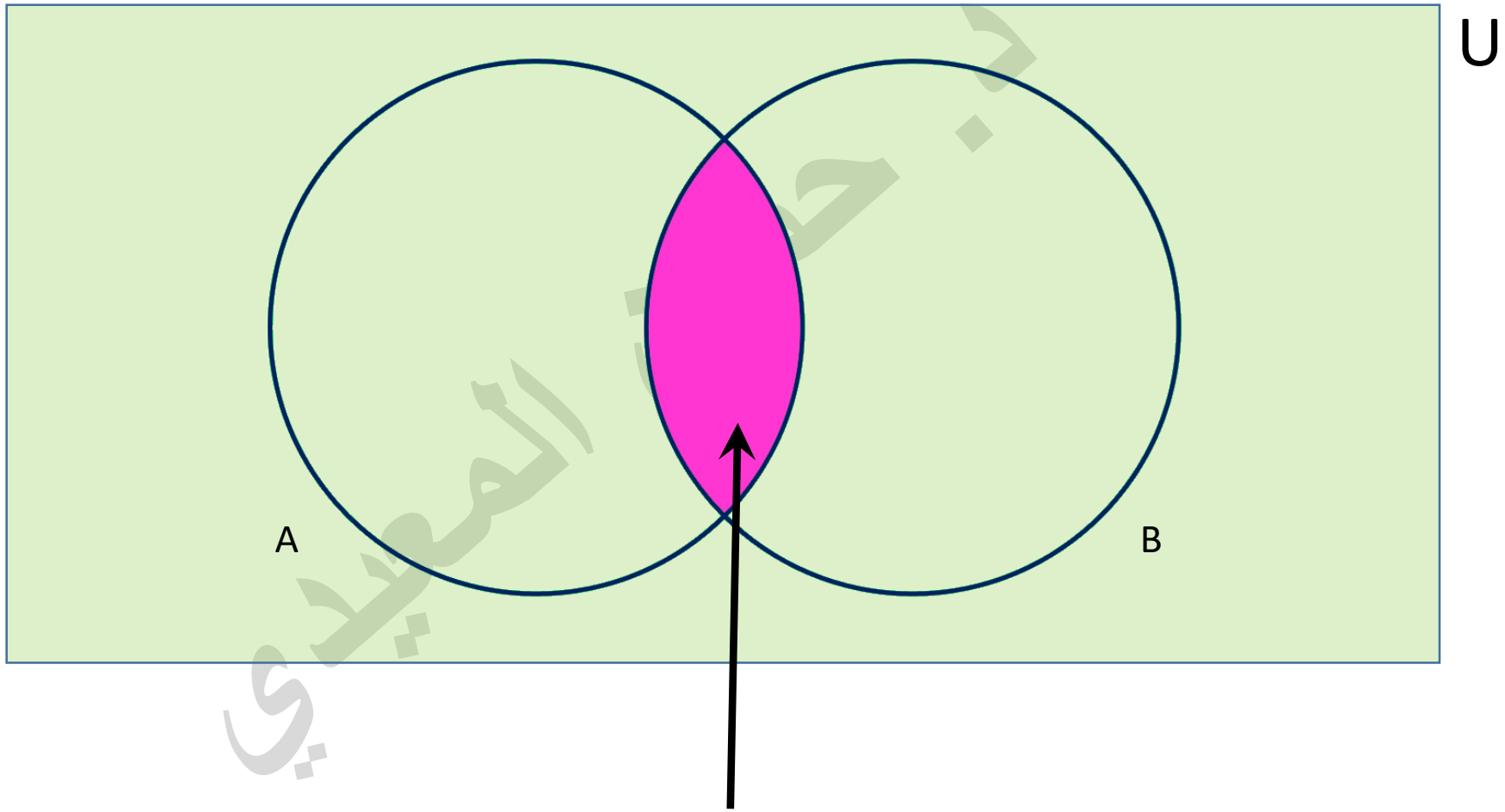
$B = \{ 3, 7, 8, 10 \}$

Find

b. $A' \cap B'$

Solution

Venn Diagram - What is the shaded region?



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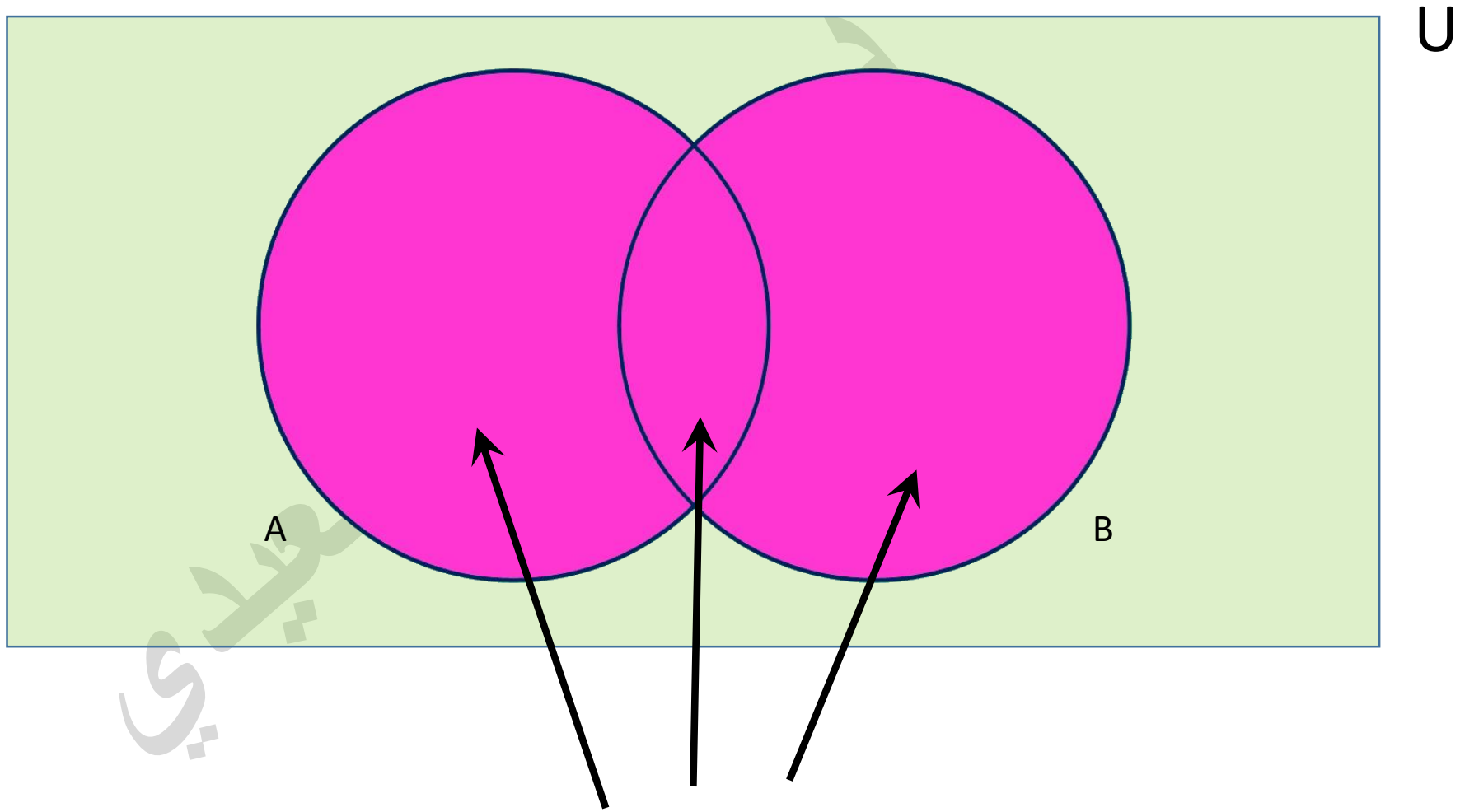

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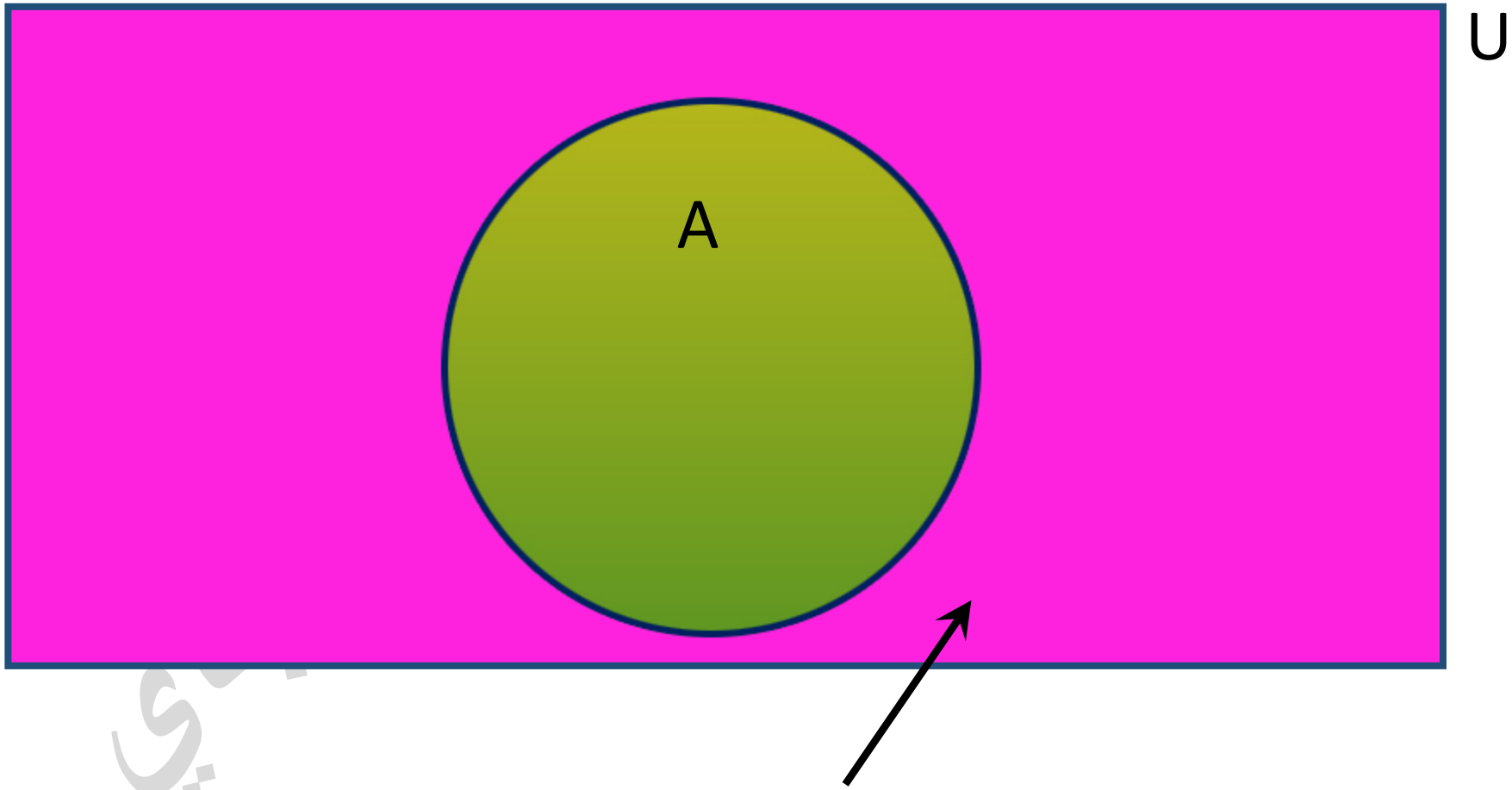

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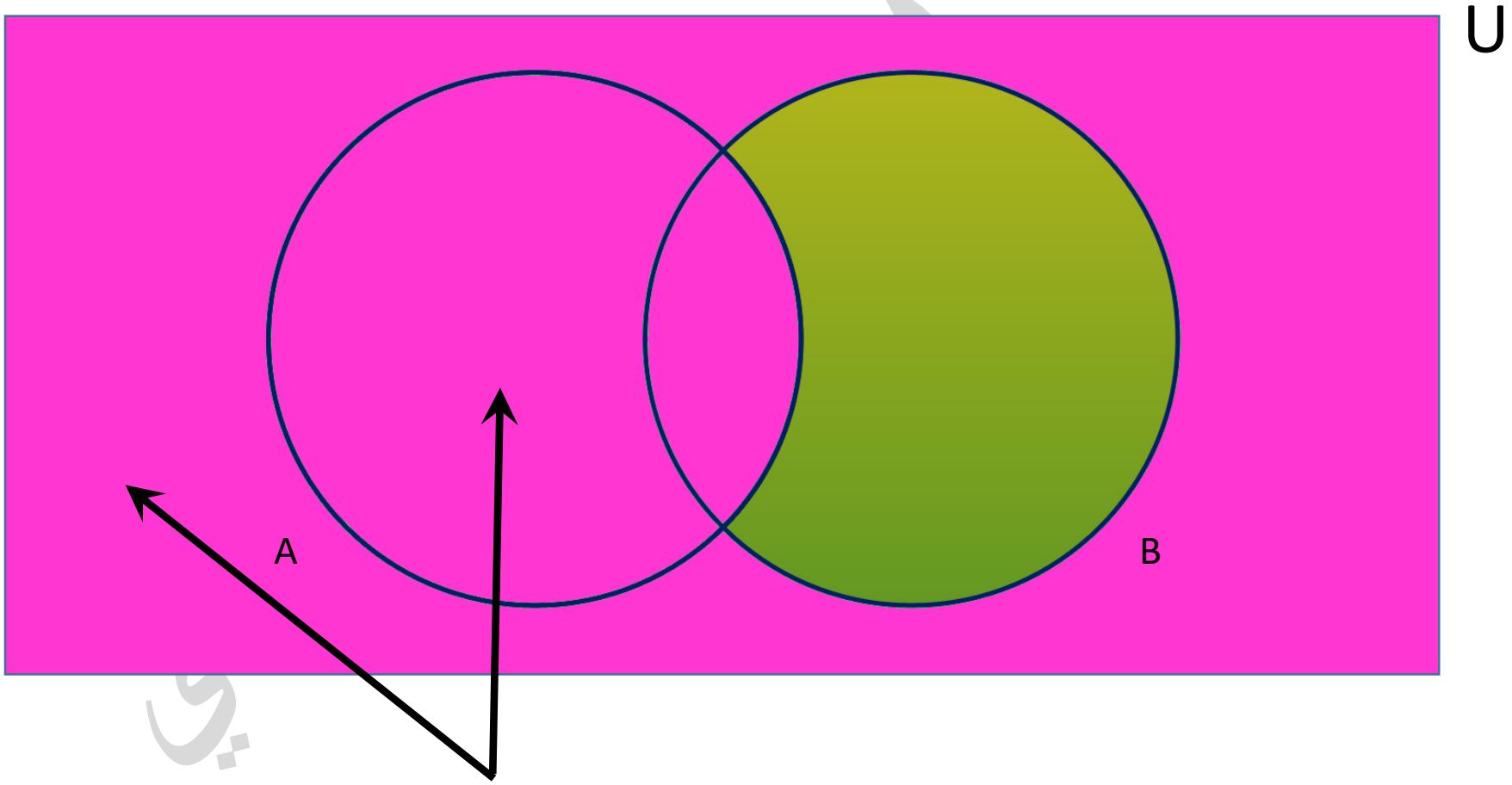

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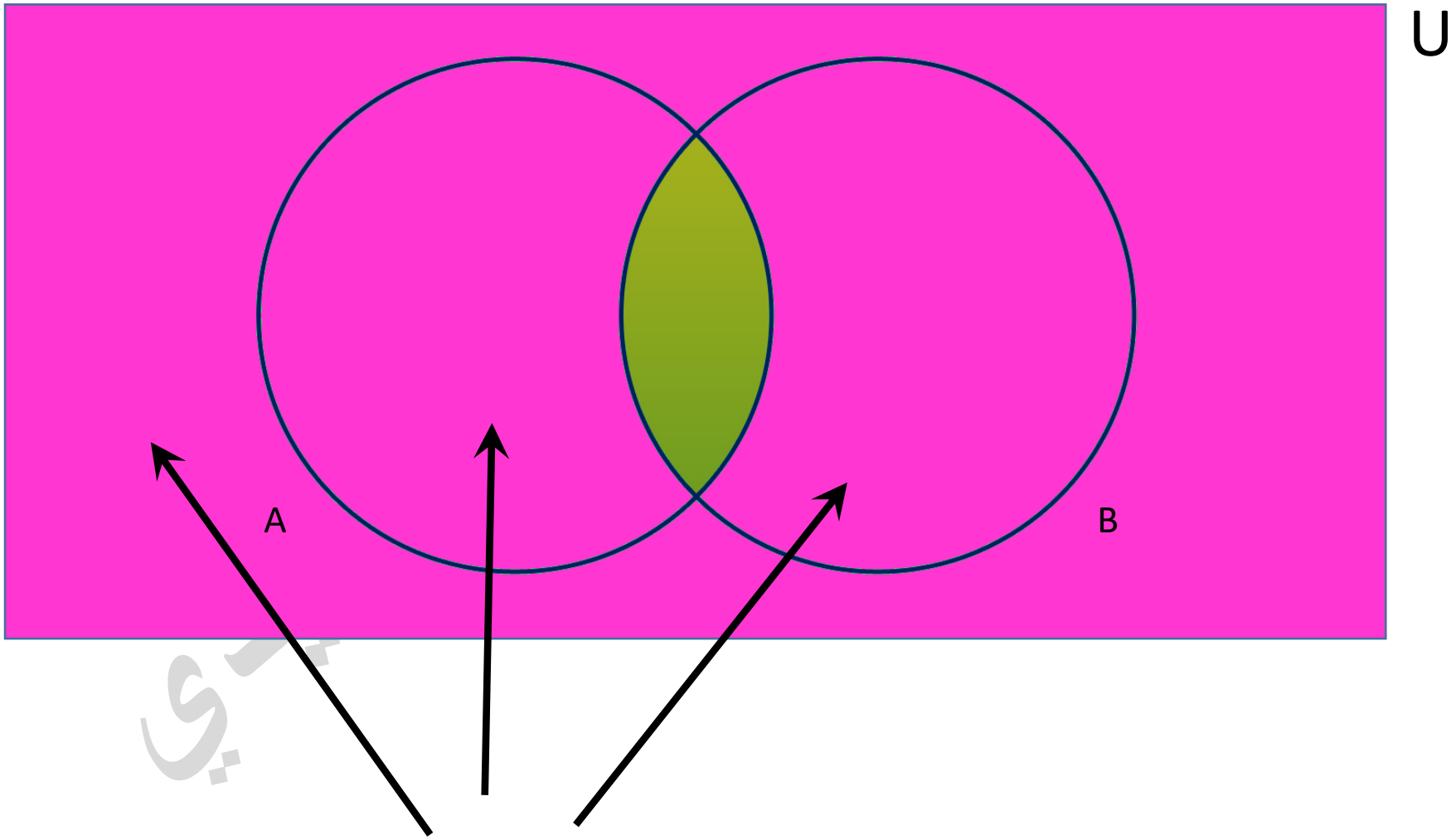

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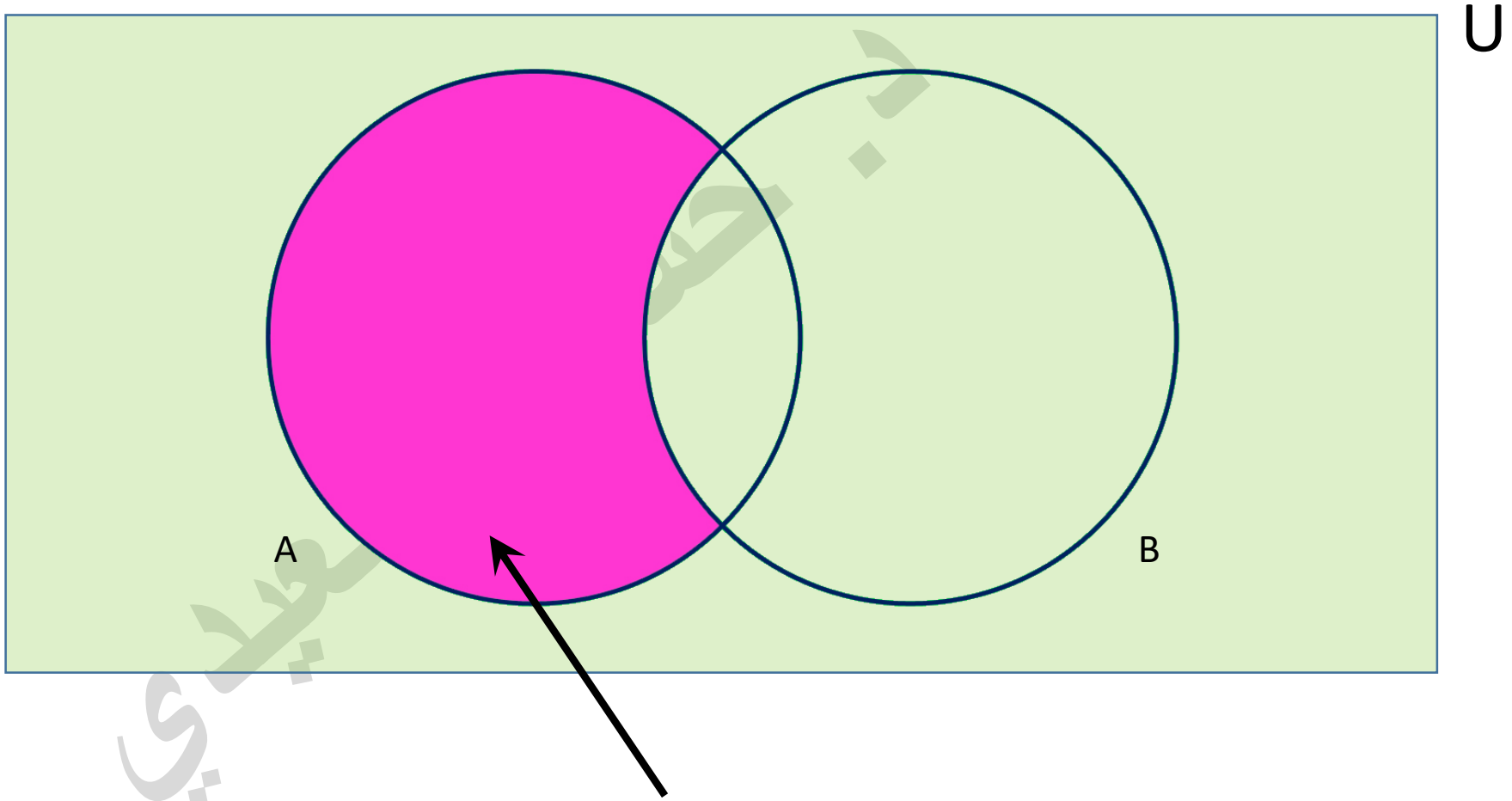
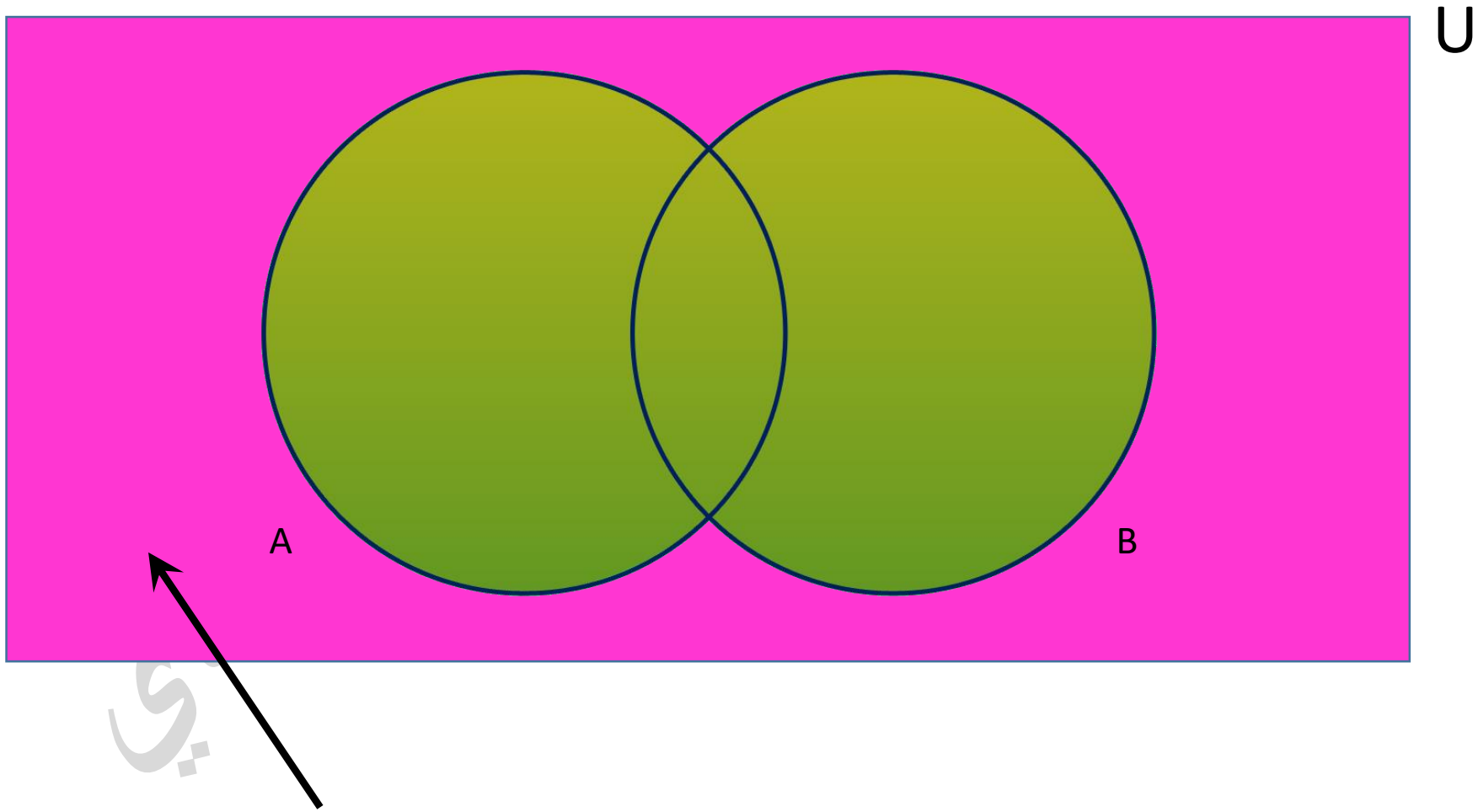
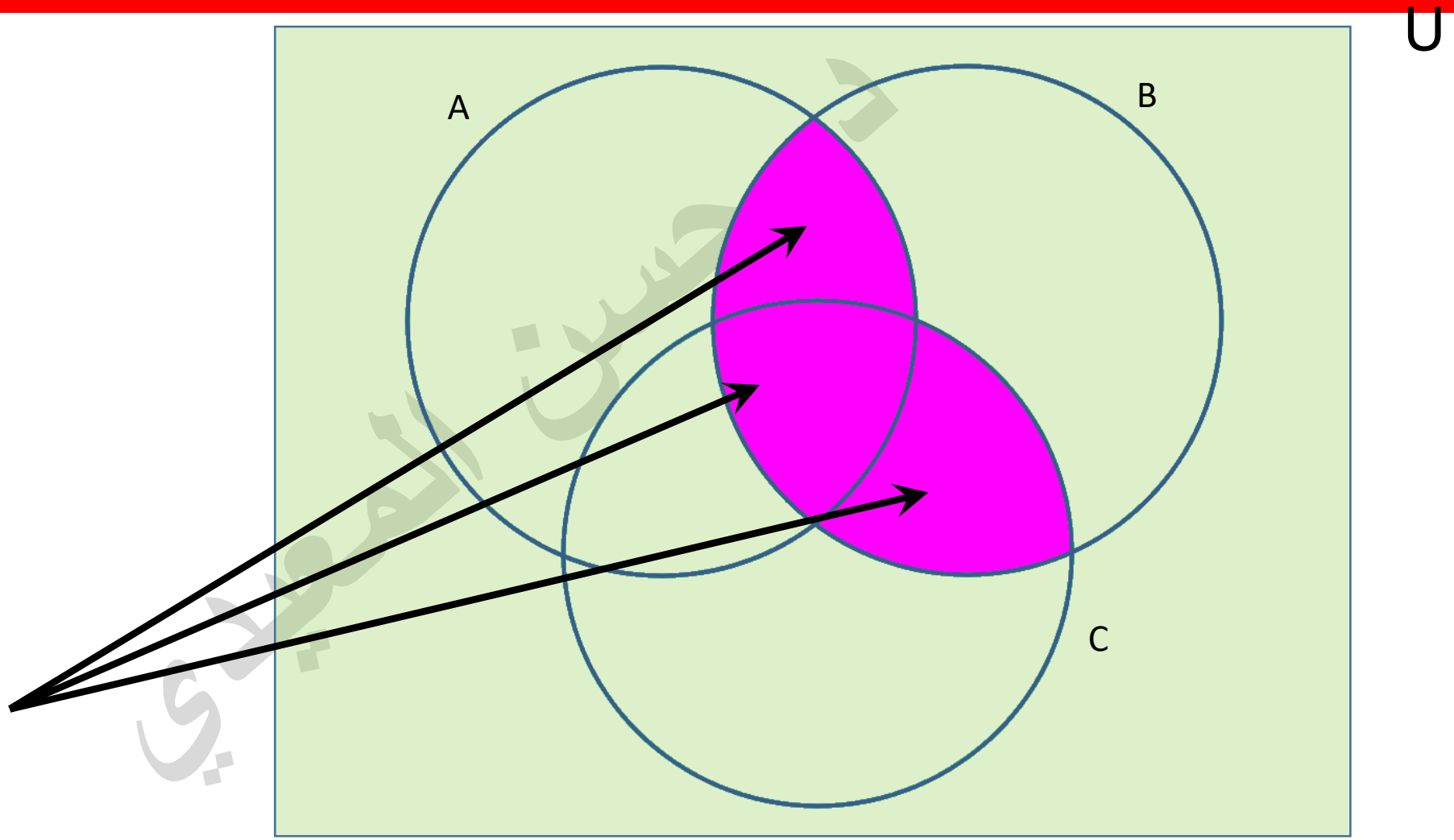


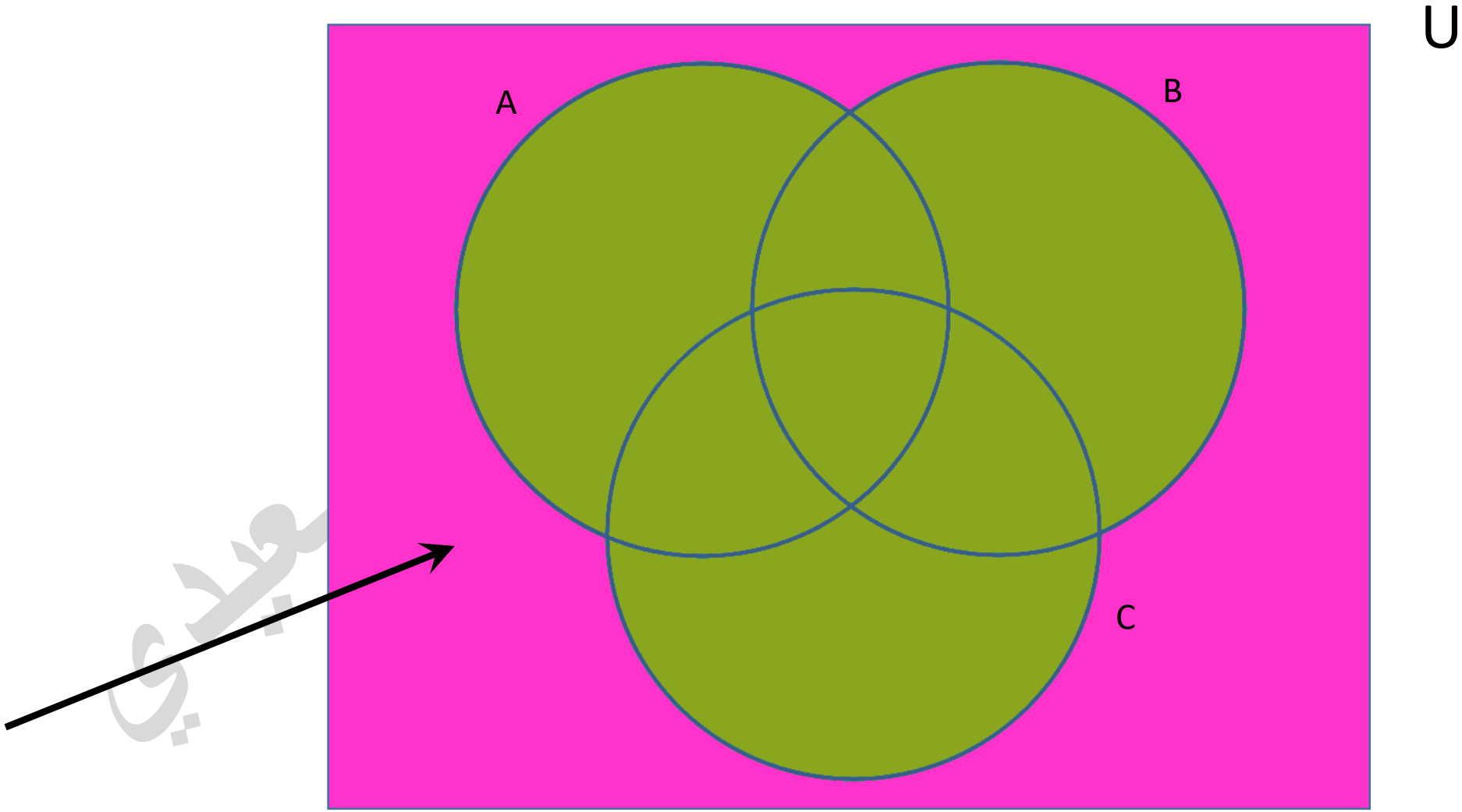
Venn Diagram - What is the shaded region?



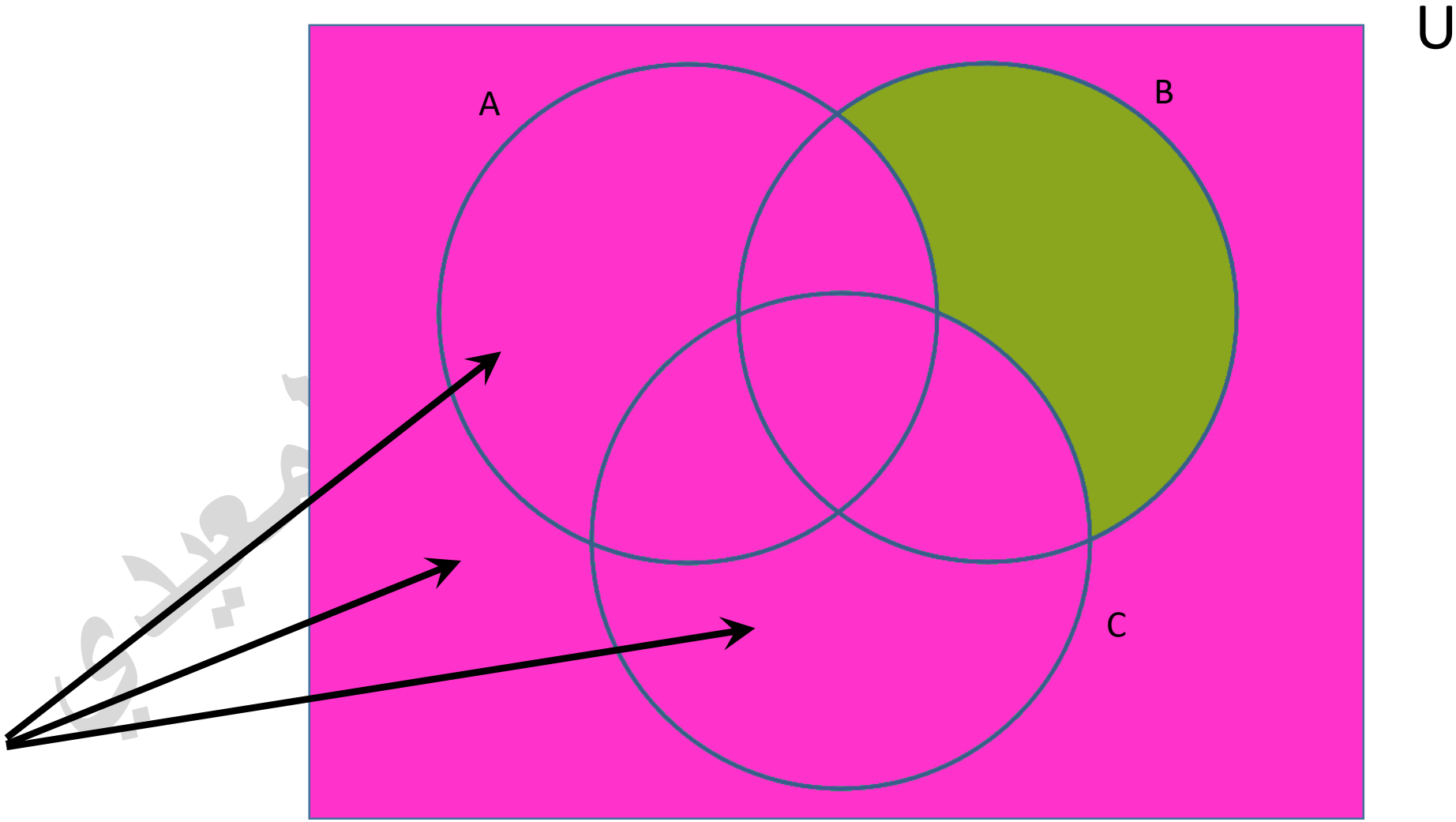
Venn Diagram - What is the shaded region?



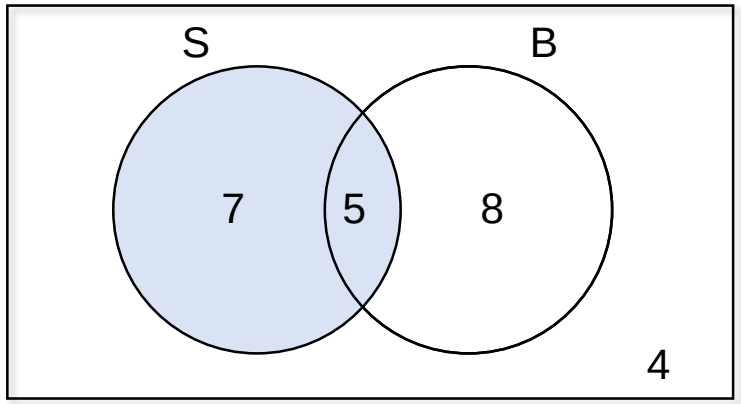
Venn Diagram - What is the shaded region?



Venn Diagram - What is the shaded region?



Review Questions



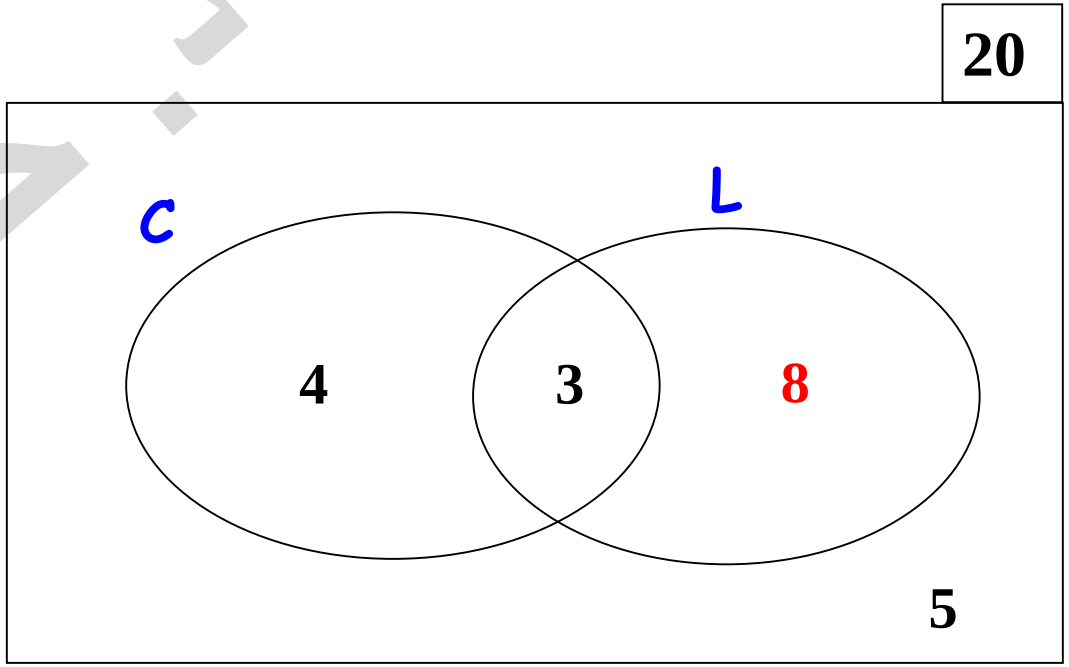
Mrs Muna asks 24 pupils in her class about their families. She sorts them into:
S - {has sisters}
B - {has brothers}
She showed her results on a Venn diagram

1	Describe the blue region in words.
2	Describe the region $S' \cup B'$ in words.
3	How many students are in S'.
4	Find $n(S \cap B)$
5	Find $\lvert S' \cap B \rvert$

Review Questions

Amongst a group of 20 students, 7 are taking C++ and of these 3 are also taking Logic. 5 are taking neither.

1	Find $\lvert C \rvert$
2	Find $\lvert L \rvert$
3	Find $\lvert C' \rvert$
4	Find $\lvert C \cap L \rvert$
5	Find $\lvert (C \cap L)' \rvert$
6	Find $\lvert C \cup L \rvert$
7	Find $\lvert (C \cup L)' \rvert$



Review Questions

1. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, and $C = \{2, 3, 5\}$.

   a. Find $A \cap B$.

   b. Find $A \cup B$.

   c. Find $A \setminus B$.

   d. Find $A \cap \overline{(B \cup C)}$.

   e. Find $A \times C$.

   f. Is $C \subseteq A$? Explain.

   g. Is $C \subseteq B$? Explain.

Review Questions

Draw a Venn diagram to represent each of the following:

a. $A \cup \overline{B}$

b. $\overline{(A \cup B)}$

c. $A \cap (B \cup C)$

d. $(A \cap B) \cup C$

e. $\overline{A} \cap B \cap \overline{C}$

f. $(A \cup B) \setminus C$

Check Chapter Learning Outcomes

CLO

1

- Represent a set using set-builder notation and roster method.
- Understand the numerical sets.
- Understand the concepts: universal set, empty set, finite and infinite Sets, set of sets.
- Describe memberships of sets, including the empty set, using proper notation, and decide whether given items are members and determine the cardinality of a given set.
- Describe the relations between sets regarding membership, equality, subset, and proper subset, using proper notation.
- Perform the operations of union, intersection, complement, and difference, complements, Cartesian products, and power sets.
- Determine whether two sets are equal by determining whether each is a subset of the other.
- Be able to draw and interpret Venn diagrams of set relations and operations and use Venn diagrams to solve problems.
- Calculate and describe the power set for small sets.
- Understand set properties.